DHRUV SINGLA

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EXPERIENCE

Operations Manager, Flo Mobility *Aug 2024 – Present*

Early operations hire at a Series-A EV fleet startup; sole owner of end-to-end ops across procurement, vendor management, and cross-functional execution.

- Redesigned core operational workflows, cutting average turnaround time by ~80% and scaling execution capacity without proportional headcount growth.
- Onboarded and managed 50+ vendors — ran RFPs, led negotiations, and drove cost optimization initiatives across procurement cycles.
- Partnered directly with engineering and product teams to translate operational pain points into fleet management product features shipped to production.
- Acted as the voice of operations in product discussions — documented pain points, wrote structured requirement briefs, and worked directly with engineering to ship fleet management features to production
- Acted as cross-functional bridge across operations, product, and engineering — a rare hybrid role at this stage of company.

EDUCATION

B.Tech, Electronics & Communication Engineering, PES University *Dec 2020 – Jun 2024*

Relevant Coursework: Machine Learning **Specialization**, Digital Image Processing, Computer Networks, Real-Time Operating Systems, Web Development, Computational Problem Solving (Python & C++)

RESEARCH PUBLICATION

Fairness in Deep Reinforcement Learning-based Joint Active and Passive Beamforming for RIS-Assisted MISO Systems | *Stack: PyTorch • CUDA • RIS Gym Environment* *July 2024*

- Co-authored peer-reviewed research applying Deep Reinforcement Learning to optimize beamforming in RIS-enhanced MISO wireless systems, achieving ~99% accuracy against theoretical benchmarks.
- Work contributed to improving signal quality and network performance in next-generation wireless infrastructure contexts.

PROJECTS

MQTT Protocol Smart Appliance Chatbot | Flask · MQTT · GPT API *Aug 2023*

- Built a Flask backend integrated with an MQTT broker enabling real-time natural language control of smart appliances — a working end-to-end IoT + LLM integration.
- GPT integration parsed user commands and mapped them to appliance actions, reducing manual interaction friction; demonstrated as a live prototype.

Collaborative Storytelling Platform | Node.js · Express · MySQL · JWT *Jun 2024*

- Full-stack web app supporting multi-user collaborative story authoring with CRUD operations, JWT authentication, MySQL persistence, and a community voting/ranking system.
- Designed and deployed the complete backend API independently, handling auth, data modeling, and vote aggregation logic.

Accelerometer-Enhanced GPS Data Logger | C · Embedded Systems *Jan 2023*

- Deployed a wireless environmental data logger collecting temperature, touch, and GPS coordinates, triggered by accelerometer events for power-efficient remote monitoring.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript (ES6+)
Web / Full-Stack: MERN Stack (MongoDB, Express, React, Node.js)
ML / Data: PyTorch, Deep Reinforcement Learning, ETL, Statistical Analysis, EDA, Power BI, Excel
Strengths: Cross-functional leadership, analytical problem-solving, data-driven decision making, startup
Tools & Platforms: Git, Linux, MATLAB, Figma, WireShark, GNS3, IoT

ADDITIONAL

- Teaching Assistant (2023): Led 8 lab sessions covering PyTorch for ML and MATLAB antenna simulation at PES University.
- Head of Discipline Committee + Head of Security + Advisor to Sponsorship Head — Maaya (PES University's flagship annual fest, ~5,000+ attendees).